

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$y + k = x + 26$

$y - k = x^2 - 5x$

In the given system of equations, k is a constant. The system has exactly one distinct real solution. What is the value of k ?

Correct Answer: 17.5, 35/2

Rationale

The correct answer is $\frac{35}{2}$. Subtracting the second equation from the first equation yields $(y + k) - (y - k) = x + 26 - (x^2 - 5x)$, or $2k = -x^2 + 6x + 26$. This is equivalent to $x^2 - 6x + (2k - 26) = 0$. It's given that the system has exactly one distinct real solution; therefore, this equation has exactly one distinct real solution. An equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, has exactly one distinct real solution when the discriminant, $b^2 - 4ac$, is equal to 0. The equation $x^2 - 6x + (2k - 26) = 0$ is of this form, where $a = 1$, $b = -6$, and $c = 2k - 26$. Substituting these values into the discriminant, $b^2 - 4ac$, yields $(-6)^2 - 4(1)(2k - 26)$. Setting the discriminant equal to 0 yields $(-6)^2 - 4(1)(2k - 26) = 0$, or $-8k + 140 = 0$. Subtracting 140 from both sides of this equation yields $-8k = -140$. Dividing both sides of this equation by -8 yields $k = \frac{35}{2}$. Note that 35/2 and 17.5 are examples of ways to enter a correct answer.